

Mwalika: Agentic AI System for eCitizen Platform

“Guiding every Kenyan through the digital nation”

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Executive Summary

Kenya’s national digital transformation relies on the eCitizen platform, which now hosts over 16,000 government services. Yet a significant digital literacy gap, highlighted by the KNBS and the Communications Authority of Kenya, prevents millions of citizens from using these services effectively. Low digital proficiency leads to abandoned applications, heavy reliance on intermediaries, increased costs, and reduced trust in government systems. This creates a structural barrier to national progress: the citizens who would benefit most from efficient, low-cost digital services are often the least able to access them.

Global peers such as Rwanda, Estonia, India, and the UAE have already integrated conversational AI into public service delivery, providing intuitive, human-centred support at scale. Without similar innovation, Kenya risks falling behind. Access to government services is not optional, it underpins entrepreneurship, social mobility, and the broader functioning of society. Digital exclusion therefore compounds inequality and weakens the nation’s development prospects.

Mwalika addresses this challenge directly. It is an agentic AI assistant embedded into eCitizen, offering natural language and voice-based guidance to help all users, regardless of digital literacy, navigate and complete government processes with ease. By providing step-by-step support, contextual explanations, and real-time assistance, Mwalika reduces friction, lowers abandonment rates, and expands meaningful access to essential public services.

The solution strongly aligns with the theme **“AI for National Prosperity.”** By widening digital inclusion, Mwalika strengthens Kenya’s institutional capacity and supports sustainable development goals. It improves economic empowerment through easier business registration, tax compliance, and access to government programmes. It advances social equity by enabling participation from citizens with low literacy,

disabilities, or limited digital experience. And it enhances service efficiency by reducing backlogs and freeing government staff to focus on complex cases.

Mwalika also contributes to **national security** by reducing dependence on cybercafé intermediaries, common points of fraud and identity misuse, and by operating on secure, self-hosted open-source AI models that keep citizen data entirely within Kenyan controlled infrastructure. This approach reinforces digital sovereignty and aligns with national cybersecurity priorities.

By increasing trust, improving accessibility, and streamlining service delivery, Mwalika helps modernise public administration and unlock wider adoption of digital services. The result is a more resilient, inclusive, and efficient public service ecosystem, one that ensures all citizens can participate fully in Kenya's digital future.

Proposed Solution

The proposed solution is an agentic AI system named **Mwalika**, designed as an intelligent virtual assistant embedded directly into the eCitizen ecosystem. The name *Mwalika* stems from the Swahili word *mwaliimu*, “teacher” or “instructor”, capturing the system's primary purpose: to guide, support, and empower citizens as they navigate digital government services. A warm, human-centred name helps demystify the technology and makes it more approachable for both novice and experienced users.

Core Functionality

Leveraging modern generative AI capabilities, Mwalika will support natural language interaction, allowing users to simply “ask” for help in plain English or Swahili. Instead of forcing citizens to memorise complex navigation paths or instructions, Mwalika translates natural queries into structured guidance. For instance, the assistant will respond to queries such as:

- “How do I update my personal information?”
- “How do I apply for a passport or driving licence?”
- “Help me register a business.”
- “How do I file my KRA tax returns?”

The agent will be capable of handling a full spectrum of queries, from basic navigation to service-specific tasks requiring multi-step workflows.

Agentic Capabilities

Unlike traditional chatbots, Mwalika will be **agentic**: capable of performing actions, not just giving instructions. With appropriate permissions and guardrails, the system could:

- Walk users through multi-step forms in real time.
- Autofill fields or pre-check information where appropriate.
- Navigate between different government service pages on behalf of users.
- Trigger form submissions or initiate service requests where legally and technically permissible.

These capabilities would be configurable, allowing different government bodies to choose which workflows Mwalika may execute independently and which require user confirmation.

Advantages of a Generative AI Assistant

Implementing a generative AI layer to the eCitizen platform offers several immediate benefits:

1. **Personalised Assistance** – Each user receives guidance tailored to their situation, literacy level, and needs.
2. **Massive Scalability** – The assistant can engage thousands of users simultaneously, relieving pressure on human support staff and reducing support backlogs.
3. **24/7 Availability** – Citizens can obtain help anytime, particularly important for low-income users who cannot access support during business hours.
4. **Reduction of Errors and Abandonment** – Step-by-step guidance significantly reduces frustration, mistakes, and abandoned applications.
5. **Inclusive Access** – Properly designed, the system can support individuals with disabilities, older adults, and users with low literacy.

To widen accessibility, Mwalika can incorporate:

- **Speech-to-text**, allowing users to speak their queries.
- **Text-to-speech**, enabling the assistant to read instructions out loud.

This is particularly valuable for:

- Users with reading or typing difficulties
- Individuals with disabilities
- Users with low literacy

Voice interfaces help transform eCitizen into a platform accessible to all, not just those comfortable with screens and keyboards.

Language and Localisation Challenges

Kenya's linguistic landscape is complex. While English and Swahili are well represented in the training datasets of models such as GPT, Gemini, Claude, and similar systems, local dialects, particularly *Sheng* and numerous indigenous languages, are underrepresented. As a result, additional **fine-tuning with Kenyan datasets** would be necessary for deeper inclusivity. Most interactions today already occur in English or Swahili, meaning Mwalika can immediately operate effectively for a majority of users, with the option to expand its linguistic capability over time.

Privacy, Security, and Data Governance

Because Mwalika is intended for government use, **data privacy and sovereignty are non-negotiable**. This requires:

- Compliance with the Data Protection Act (2019)
- Secure handling of personally identifiable information (PII)
- Clear user consent for data usage
- Transparent data flows and logging
- Strict guardrails preventing unauthorised data access

The AI layer must operate with **open-source, self-hostable models**, ensuring that citizen data does not leave government-controlled environments. Suitable models include:

- OpenAI's OpenOSS models
- Meta's LLaMA family
- Mistral models

These models can be fine-tuned further to increase their effectiveness in local contexts. Overall, this approach ensures performance while maintaining full control over sensitive data.

Proposed Architecture

The diagram below presents a high-level system architecture for integrating Mwalika as an adjacent service to the eCitizen platform and its supporting government systems. It is not intended to represent a full production architecture; rather, it highlights the core components, data flows, and security boundaries required to deploy Mwalika effectively. The purpose of this simplified view is to illustrate how the AI layer, application logic, and government service integrations work together within a secure compute environment to deliver guided, intelligent assistance to citizens using eCitizen.

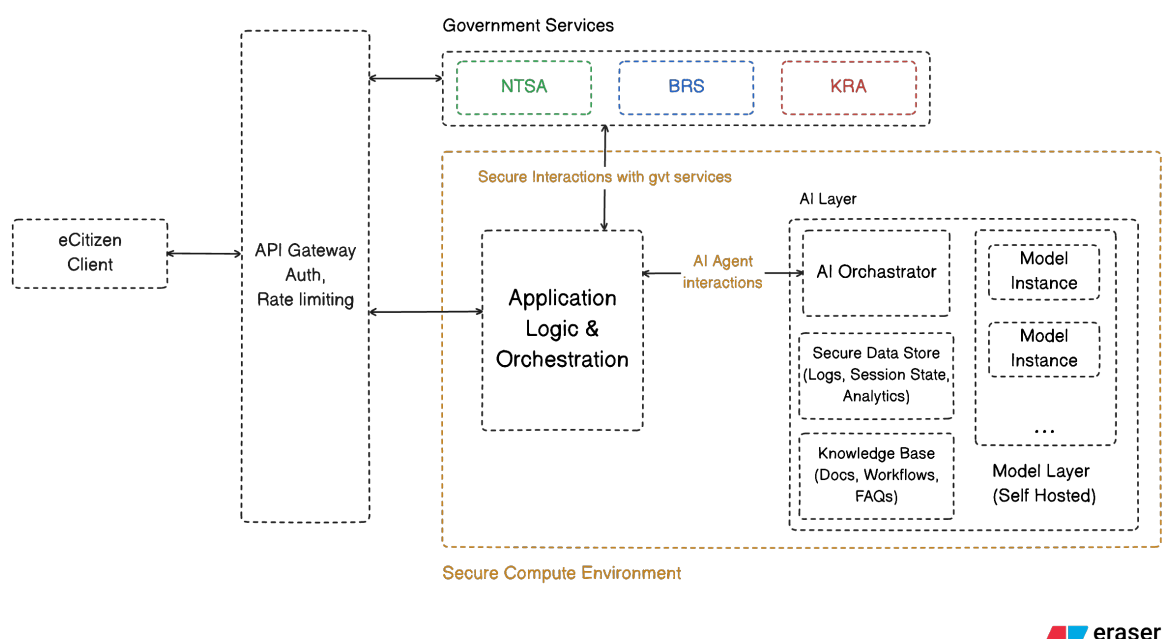


FIGURE 1: HIGH-LEVEL ARCHITECTURE OF THE MWALIKA AGENTIC AI SYSTEM

Conclusion

Ultimately, Mwalika is more than a chatbot. It is a digital guide, a teacher, helping Kenyan citizens navigate government services confidently and independently. By integrating an open-source, secure, agentic AI assistant into eCitizen, Kenya can deliver a more inclusive, efficient, and humane digital public service experience. This innovation directly supports national development by ensuring that no citizen is left behind in the country's digital transition.